

TeX math sampler for format: typst

1. Basics: sub/superscripts, fractions, roots

Inline: x^2 , x_i , x_i^2 , $\frac{a}{b}$, $\sqrt{x}$, $\sqrt[3]{x}$, $\frac{1}{2}$, $\frac{1}{2}$.

2. Decorated symbols

$\bar{x}$, $\hat{x}$, $\tilde{x}$, $\vec{x}$, $\dot{x}$, $\ddot{x}$, $\overline{xyz}$, $\widehat{xyz}$, $\widetilde{xyz}$, $\overrightarrow{AB}$, $\overbrace{x+y+z}^n$, $\underbrace{a+b+c}_m$.

3. Greek letters

$\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega$

Capitals: $\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Upsilon, \Phi, \Psi, \Omega$

4. Operators

$\sum_{i=1}^n x_i$, $\prod_{i=1}^n x_i$, $\int_0^1 f(x) dx$, $\iint_D$, $\iiint_V$, $\oint$, $\lim_{x \rightarrow \infty}$, $\limsup$, $\liminf$, $\min_x$, $\max_x$.

5. Relations

$a = b$, $a \neq b$, $a \leq b$, $a \geq b$, $a \equiv b$, $a \approx b$, $a \sim b$, $a \simeq b$, $a \cong b$, $a \propto b$, $a \ll b$, $a \gg b$, $a \subset b$, $a \supseteq b$, $a \in b$, $a \notin b$.

6. aligned (amsmath)

$$\begin{aligned} f(x) &= (x+1)^2 \\ &= x^2 + 2x + 1 \end{aligned}$$

7. gather

$$\begin{aligned} a &= b + c \\ d &= e - f \end{aligned}$$

8. cases

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

9. Matrices

`pmatrix` and `bmatrix` work. `vmatrix` and `Vmatrix` are broken — see the “Known broken” section below.

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

10. Delimiters

$\left(\frac{a}{b}\right)$, $[x]$, $\{y\}$, $|z|$, $\lfloor x \rfloor$, $\lceil x \rceil$, $\langle \phi \rangle$.

11. Text and font styles

for all $x \in X$, `rm`, `bf`, `it`, `sf`, `tt`, $\mathcal{AL}$, $\mathbb{RN}$, $\mathfrak{RN}$.

12. Spacing

$a\,b$ (thin), $a\,b$ (medium), ab (thick) $b\neg$, $a\quad b$, $a\qquad b$.

13. Arrows

$\rightarrow$, $\leftarrow$, $\rightrightarrows$, $\leftrightsquigarrow$, $\Rightarrow$, $\Leftarrow$, $\Leftrightarrow$, $\mapsto$, $\overset{f}{\rightarrow}$, $\overset{g}{\leftarrow}$, $\overset{!}{=}$, $\lim_{n \rightarrow \infty}$.

14. Substack and stacked limits

$$\sum_{\substack{i=1 \\ i \neq j}}^n a_i$$

15. mathtools (partial)

`\coloneqq` and `\eqqcolon` work; most other mathtools commands don't (see "Known broken" below).

$a := b$, $a =: b$.

Known broken

The following constructs either crash the Typst compile, cause Pandoc to warn and pass raw TeX through, or silently drop content. Each is commented out to keep the rest of this file renderable.

`\matrix` and `\Vmatrix` (fatal Typst error)

Pandoc's Typst writer emits `mat(delim: "||")` but Typst requires a single-character delim. Workaround: raw Typst, or `\pmatrix` with explicit `\lvert`/`\rvert` delimiters.

`\mathclap` (passes through as literal TeX)

Not recognised by texmath; the source appears verbatim in the PDF.

`\begin{dcases}` (Pandoc parse error)

Not recognised by texmath. Use plain `cases`.

`\tag{label}` (silently dropped)

The tag never appears in the output.

`\color{red}{...}` (silently dropped)

`\color` is ignored; content renders in the default color.

```tikz` blocks (not converted)

The raw `\begin{tikzpicture}...\end{tikzpicture}` source appears as a code block in the PDF, not a diagram.